26900556 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEZGY302 - CLIMATE CHANGE AND DISASTER RISK REDUCTION

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which of the following is a phenological change due to climate change? [U] / [CO1]

   a). Habitat loss and fragmentation

   b). Increased risk of extinction

   c). Water and food scarcity

   d). Migratory birds arrive at their breeding grounds after food resources are depleted

2. Burning of fossil fuels emit large amounts of ---------- into the atmosphere. [U] / [CO1]

   a). Nitrogen b). Oxygen

   c). Carbon dioxide d). Ammonia

3. Climate change can affect rural livelihoods mainly through changes in _____. [U] / [CO2]

a). Temperature and rainfall b). Building height

c). Traffic density d). Industrial production

4. Which group is considerd highly vulnerable during disasters? [K] / [CO2]

a). Women and children b). Elderly persons

c). Persons with disabilities d). All of the above

5. Situation analysis assess [K] / [CO3]

a). Only human-induced hazards b). Only natural hazards

c). Both natural and human-induced hazards d). Industrial hazards

6. SDMA stands for [K] / [CO3]

a). State Disaster Management Authority b). State Disaster Mitigation Association

c). State Development Management Agency d). Social Disaster Management Authority

7. The main aim of IWRM is the management of water resources in a: [K] / [CO4]

a). Commercial and unrestricted manner b). Completely private manner

c). Sustainable, equitable and efficient manner d). Temporary manner

8. Which ecosystem helps reduce coastal erosion and storm surge impacts? [K] / [CO4]

a). Grasslands b). Mangroves

c). Agricultural fields d). Deserts

9. Which of the following is a measure for strengthening community resilience? [U] / [CO4]

a). Disaster awareness programmes b). Early warning systems

c). Community-based disaster management d). All of the above

10. Which event is discussed as a successful disaster-management case in Kerala? [K] / [CO5]

a). Kerala Floods 2018 b). Kerala earthquake 2000

c). Kerala tsunami 2010 d). Kerala drought 1990

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. Differentiate physical vulnerability from social vulnerability. [U] / [CO3]

12. What are earthquakes? Mention the major impacts of land and submarine earthquakes. [U] / [CO1]

13. What is meant by carbon neutral? [K] / [CO1]

14. Define fog. [K] / [CO1]

15. Name four disease outbreaks following floods. [U] / [CO2]

16. Mention two psycho-social impacts of water scarcity. [K] / [CO2]

17. Define livelihood resilience. [K] / [CO2]

18. Mention the three water-security challenges faced by Kerala. [K] / [CO2]

19. Explain various components of disaster risk management. [U] / [CO3]

20. Summarize the responsibilities of local authorities during a disaster. [U] / [CO4]

21. What is risk knowledge and why is it important in an Early Warning System? [U] / [CO5]

22. What is the importance of community participation in climate resilience? [A] / [CO5]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Explain the concept of green auditing. [U] / [CO1]

24. Explain the psycho-social impacts of climate change in Kerala and India. [U] / [CO2]

25. Explain in detail the tools and technologies used in Disaster Risk Management. [U] / [CO3]

26. Describe how disaster management planning is done at various levels of governance. [U] / [CO3]

27. Explain the importance of land-use planning in disaster risk reduction. [U] / [CO3]

28. Explain the major climate-related hazards that require CCA–DRR strategies. [U] / [CO4]

29. Discuss how the field of geo-informatics can contribute towards Disaster Management. [A] / [CO4]

30. Discuss the major international policies and agreements developed to address climate change. [K] / [CO5]

[5x6 = 30]